Decomposing Contextual Embeddings: Mechanistic Interpretability via Sparse Autoencoders

Danil Fathutdinov

Innopolis University | Natural Language Processing | Spring 2026

1. Motivation & Research Question

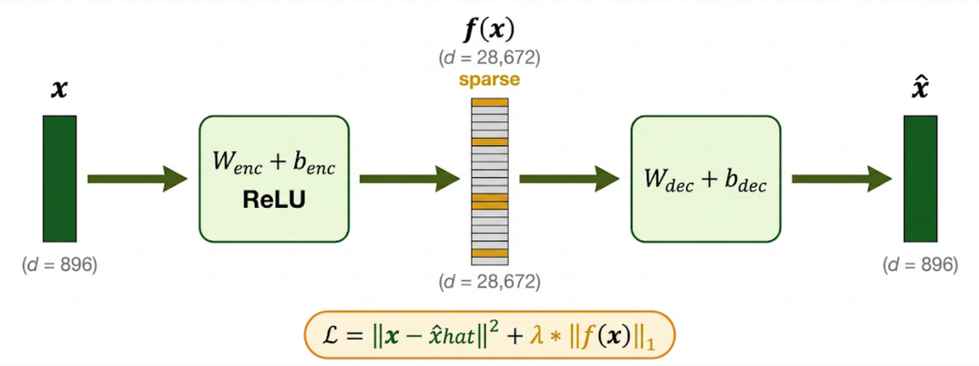
LLMs encode knowledge in dense vectors where single neurons respond to **multiple unrelated concepts** (**polysemanticity**), obscuring internal reasoning and blocking reliable model auditing.

Can we decompose polysemantic activations into sparse monosemantic features and causally steer generation?

This study trains **Sparse Autoencoders** on residual stream activations at three transformer depths, then validate features via activation steering following Anthropic's "Towards Monosemanticity" (2023).

2. SAE Architecture

The SAE maps $x \in R^{896}$ into a 32× **overcomplete** latent space, then reconstructs from a sparse subset of active features.



Stage	Formulation
Encode	$f(x) = \mathrm{ReLU}\big((x - b_{\mathrm{dec}})\, W_{\mathrm{enc}} + b_{\mathrm{enc}}\big)$
Decode	$\hat{x} = f(x)\, W_{\mathrm{dec}} + b_{\mathrm{dec}}$
Loss	$\mathcal{L} = \lVert x - \hat{x} \rVert_2^2 + \lambda \lVert f(x) \rVert_1$
Constraint	$\lVert W_{\mathrm{dec}}[i] \rVert_2 = 1 \;\; \forall i$

Key design choices: pre-centering by b_{dec}, unit-norm decoder rows to prevent degenerate scaling, L1 warmup over 200 steps to avoid early feature collapse.

3. Activation Steering Protocol

This study injects a scaled decoder direction into the residual stream during the forward pass to test **causal significance**:

$$\mathbf{h}_\ell \leftarrow \mathbf{h}_\ell + \alpha \cdot W_{\mathrm{dec}}[i]$$

α	Expected Effect
$\alpha \gg 0$	Amplify the target concept
$\alpha = 0$	Unmodified baseline
$\alpha \ll 0$	Suppress the target concept

We sweep $\alpha \in [-10, +10]$ and track both perplexity shift and qualitative output changes.

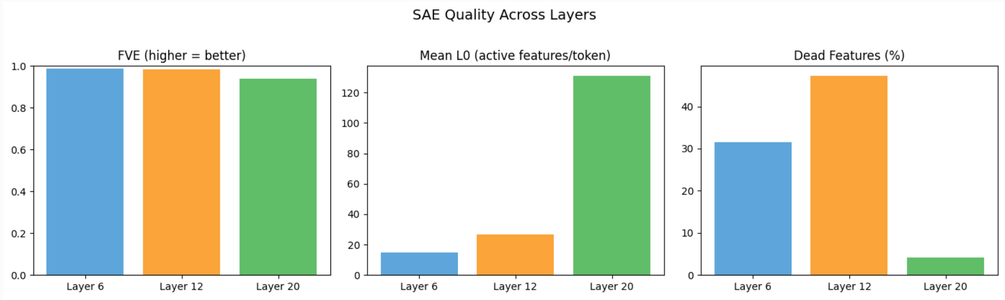
4. Experimental Setup

Parameter	Value
Base Model	Qwen2.5-0.5B (24 layers, d=896)
Dataset	FineWeb-Edu (sample-10BT)
Tokens collected	2M per layer
SAE dictionary	$896 \times 32 = 28{,}672$ features
Optimizer	Adam, lr $= 10^{-3}$
Sparsity	$\lambda = 3.5 \times 10^{-3}$ (post warmup)
Target layers	6 (early), 12 (middle), 20 (late)
Interpretation	500 texts, top-10 contexts per feature

Early layers capture surface patterns; middle layers encode semantics; late layers specialize in next-token prediction. Comparing all three reveals how representations transform across depth.

5. Cross-Layer Reconstruction Quality

Layer	FVE ↑	L0 ↓	Dead %	Alive	Feature Type
6 (Early)	**0.989**	**14.9**	31.6	19,610	Surface / genre patterns
12 (Mid)	0.983	26.6	47.3	15,122	**Semantic / conceptual**
20 (Late)	0.939	131.0	**4.2**	27,470	Subword predictive



Early layers decompose into ∼15 sparse features per token; late layers need ∼131 yet explain less variance, confirming deeper representations are more distributed.

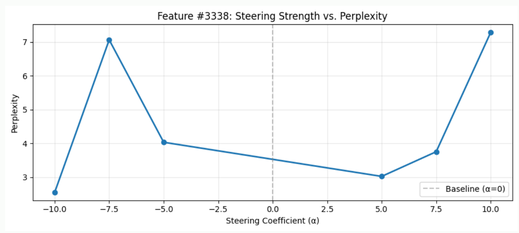
6. Monosemantic Features (Layer 12)

ID	Concept	Top Activating Context
#21836	Statistical reporting	"accounting for **18%** of global output"
#3338	Speech attribution	"increases the motivation,**" said** Max…"
#8966	Exemplification	"factors, **such as** financial hardship…"
#16902	Concessive discourse	"in the construction. **Even** after…"
#11694	Academic titles	"Biochemical Zoology, **Dr** Euan…"

Each feature fires consistently on one coherent pattern across hundreds of documents, confirming monosemanticity. Layer 6 features encode surface patterns (news headers), while layer 20 features fire on subword prefixes ("Re-", "con-").

7. Steering Case Study: Feature #3338 (Layer 12)

Prompt: "Now, let's talk about clothes and how to style the outfit for date night."



Perplexity vs. α for feature #3338. Moderate values alter style; extremes cause degeneration.

α	PPL	Observed Behavior
−10	2.55	Repetitive questions; attribution gone
−5	4.03	Casual advice; no third-party voices
0	**3.3**	**Baseline: balanced paragraph**
+5	3.02	Numbered tips; formal tone
+7.5	3.75	Fabricated: *"says Andrea"*
+10	7.28	Pure citation cascade

Causal Validation: Smooth transition from suppression through a useful range ($|\alpha| \approx 5$) to semantic takeover confirms Feature #3338 is a genuine computational unit controlling attribution style.

8. Conclusions

1. **Monosemanticity achieved.** SAEs decompose polysemantic activations into single-concept directions verified across diverse documents.
2. **Middle layers are optimal.** Layer 12 yields the richest semantic features and most effective steering.
3. **Causal validity confirmed.** Steering produces predictable, graded changes with a clear sweet spot, proving features are computational units.
4. **Practical value.** SAE steering enables training-free controllable generation for safety and style control.

Limitations: 2M tokens suffices for proof of concept; production SAEs need billions. The 47.3% dead rate at layer 12 would decrease with more data and larger dictionaries (64x+). Future work: automated labeling via LLM judges, multi-epoch training, and extension to instruction-tuned models.